

MTH645

Final-Term Important

Objective

- 1) The number of most widely used distance formulae between two fuzzy sets : (ans) 4
- 2) The max(min) distance formula b/w two fuzzy sets is: (ans) $d(A, B) = \max[\min(\mu_A(x_i), \mu_B(x_i))]$
- 3) Let $A(x, 0.4)$ and $B(x, 0.7)$ be two fuzzy sets on $X = \{x_1\}$, max(min) distance b/w A and B is: (ans) 0.4
- 4) If $A(x, 0.2)$ and $B(x, 0.5)$ are two fuzzy sets Then $\max(A, B) = \underline{\hspace{1cm}}$. (ans) 0.5
- 5) If $A(x, 0.4)$ and $B(x, 0.8)$ are two fuzzy sets, then $\min(A, B) = \underline{\hspace{1cm}}$. (ans) 0.4
- 6) Let $X = \{x_1, x_2, \dots, x_n\}$ be a universal set and A, B are two fuzzy sets on X , the distance $d(A, B)$ satisfy the condition:
(ans) $d(A, B) = 0$ if and only if $A = B$.
- 8) A fuzzy set has two membership degrees is called Intuitionistic. (ans) Intuitionistic
- 9) A fuzzy set which has three elements whose sum always lies b/w $[0, 1]$ membership degree is called Picture fuzzy Set. (ans) Picture fuzzy Set

(10) Let $A = \{ \langle x, (\mu_A(x), \gamma_A(x)) \rangle \}$ and $B = \{ \langle x, (\mu_B(x), \gamma_B(x)) \rangle \}$ be two intuitionistic fuzzy sets on X then $A=B$ if and only if _____?

(ans) $\mu_A(x) = \mu_B(x)$ and $\gamma_A(x) = \gamma_B(x)$

(11) Let $X = \{x\}$ be a fixed set and $A = \{ \langle \mu_A(x), \gamma_A(x), \beta_A(x) \rangle \}$ and $B = \{ \langle \mu_B(x), \gamma_B(x), \beta_B(x) \rangle \}$ are two picture fuzzy sets on X then $A \leq B$ if and only if

(ans) $\mu_A(x) \leq \mu_B(x)$ and $\gamma_A(x) \leq \gamma_B(x)$ and $\beta_A(x) \geq \beta_B(x)$

(12) If $P = (\alpha, \gamma, \beta)$, is a PFS, then $\beta =$ _____?

(ans) $1 - \alpha - \gamma$

(13) If $P = (\alpha, ?, \beta)$ is a PFS, then $\gamma =$ _____?

$\gamma = 1 - \alpha - \beta$

(14) The missing element of PFS $(-, 0.2, 0.35)$ is

(ans) 0.45

(15) The missing element of a PFS $(0.45, -, 0.55)$ is _____. (ans) 0

(16) A fuzzy set $F = \{ \langle x, \{0.2, 0.3\} \rangle \}$ is named as: (ans) Hesitant fuzzy set

(17) A hesitant fuzzy set is written as $H = \{ \langle x, h(x) \rangle \mid \text{for all } x \text{ in } X \}$ where $h(x)$ is a set of some distinct value belong to _____.

(ans) $[0, 1]$

18) For any hesitant fuzzy set $h(x)$ and $\gamma \in h(x)$
then lower bound $= h^-(x) = \underline{\hspace{2cm}} ?$

(ans) $h^-(x) = \min h(x)$

19) For any hesitant fuzzy set $h(x)$ and $\gamma \in h(x)$
then upper bound $= h^+(x) = \underline{\hspace{2cm}} ?$

(ans) $h^+(x) = \max h(x)$

20) The hamming distance b/w two HFS
 $h_1 = \{x_1, (0.3)\}$ and $h_2 = \{x_1, (0.5)\}$ is $\underline{\hspace{2cm}} ?$

(ans) 0.2

21) The Euclidean distance b/w two HFS
 $h_1 = \{x_1, (0.4)\}$ and $h_2 = \{x_2, (0.7)\}$ is $\underline{\hspace{2cm}} ?$

(ans) 0.3

22) let Δ is a t-norm, then $0 \Delta 0 = \underline{\hspace{2cm}} ?$

(ans) 0

23) if $f(x) = e^{\frac{x}{1-x}}$, then $a = \underline{\hspace{2cm}} ?$

(ans) 0

24) If $f(x) = e^{x-5}$, then $f^{-1}(x) = \underline{\hspace{2cm}} ?$

(ans) $5 + \ln x$

25) $R \circ R \leq R$ iff $\underline{\hspace{2cm}} ?$ (ans) R is Transitive

26) ∇ -implication implies $(x \Rightarrow y)$ is $\underline{\hspace{2cm}} ?$

(ans) $(1-x) \nabla y$

27) If $f(x) = e^{x-10}$, then $f^{-1}(x) = \underline{\hspace{2cm}} ?$

(ans) $10 + \ln x$

28) If $h = \{0.6\}$ is HFS and $\lambda = 2$ then the value $\lambda h =$ — ?

(ans) 0.84

29) If $h = \{0.3\}$ is a HFS and $\lambda = 2$ then the value of $h^\lambda =$ — ?

(ans) 0.09

30) Two object are said to be similar if both have — ? (ans) Same Shape.

31) the process of finding the most resemblance b/w two object is called — ?

(ans) Similarity measure.

32) Pick up intuitionistic fuzzy set

(ans) $\{(x, (0.5, 0.4))\}$

33) Let Δ_1 is a t-norm then $0.5 \Delta_1 0.8 =$ — ?

(ans) 0.3

34) If Δ is archimedeana then $\forall a, b \in (0, 1)$ — ?

(ans) $a \Delta b < b$

35) If $A = \{(u_A(x), \cancel{u}_A(x), \beta_A(x))\}$ is a Picture fuzzy set on X then A^c is

(ans) $\{\beta_A(x), \gamma_A(x), u_A(x)\}$

36) Fuzzy implication 0 implies 1 is = — ?

(ans) 1

37) A fuzzy binary relation R is Reflexive if — ? (ans) $R(x, x) = 1$